

Contextual Inquiry (Observation) Protocol

Standalone extract of the Phase 1 report

CSE 347/652 | DES 306/525 Usable Security and Privacy
IIIT-Delhi · Monsoon 2026 · Instructor: Dr. Arun Balaji Buduru

Phase 1: User Studies to evolve Requirements, Design and Workflows
Submission: 4 October 2026

Team: [INSERT TEAM MEMBER NAMES AND ROLL NUMBERS]

Primary research status: instruments complete; fieldwork data not yet collected. All placeholders are marked [DATA REQUIRED]. No primary data has been fabricated.

Appendix D. Contextual Inquiry (Observation) Protocol

Purpose. Establish ground truth for how entry actually happens: step count, timing, queue behaviour, guard actions, exceptions, and environmental constraints. Interviews tell us what people believe they do; observation shows what happens.

Type. Non-participant, non-interfering observation in a public area. Observers do not speak to people in the queue, do not handle cards, do not photograph or film, and do not ask guards questions during the session (questions go to the interview).

D.1 Pre-conditions

- Written or e-mail permission from the security supervisor; inform the guards on shift before starting.
- Observers wear visible college IDs and stand where the supervisor agrees, out of the flow.
- Stop immediately if asked by security staff or if an incident occurs.

D.2 Sessions

Session	Target window	Rationale
S1	Morning peak (e.g., 08:30–09:15)	Maximum load; queue formation; speed pressure
S2	Off-peak (e.g., 14:30–15:15)	Baseline behaviour without pressure
S3	Evening / hostel return (e.g., 21:30–22:15)	Low light; different guard shift; visitor sign-out
S4 (optional)	Second gate, any window	Compare gates

Two observers per session: Observer A times individuals; Observer B writes field notes and counts queue length every 2 minutes.

D.3 Timing method

- Use a stopwatch app with lap function.
- T0: person reaches the guard's position (front of queue). T1: person passes the gate line or is turned away. Record T1–T0 in seconds.
- Classify each timed person as: U (uncontested: card shown or not, no stop), Q (questioned: guard stops and speaks), N (no card visible), V (visitor), R (refused).
- Time at least 30 consecutive people per session, or the whole session if fewer.
- Record queue length (number waiting behind the person at the guard) at 2-minute intervals.

D.4 Observation checklist (Observer B)

Environment

- ☐ Lighting (adequate / dim / glare)
- ☐ Weather / shelter for guard and queue
- ☐ Noise level
- ☐ Position of guard relative to flow; is there a desk, booth, barrier, turnstile?
- ☐ Power outlet / network equipment visible?
- ☐ Space available for a reader or display; would a queue block it?

Process

- ☐ How does the guard signal "go ahead"? (nod, gesture, verbal, nothing)
- ☐ Where is the card when shown? (in hand, on lanyard, waved from distance, not shown)
- ☐ Does the guard ever look at the expiry date? Turn the card over?
- ☐ Does the guard compare photo to face? For how long?
- ☐ Does behaviour change when the queue grows? At what length?
- ☐ Are groups checked individually or as a group?
- ☐ Vehicle gate: what is checked and from what distance?

Exceptions (record each occurrence with time, type, duration, outcome)

- ☐ No card
- ☐ Guard questions holder
- ☐ Guard calls someone
- ☐ Refusal
- ☐ Visitor registration: steps, what is written, who writes, time taken, where the register sits and who can read it
- ☐ Any apparent workaround (e.g., friend passing card back, guard waving group)

Guard workload

- ☐ Concurrent duties (phone, vehicle barrier, register, radio)
- ☐ Number of guards present
- ☐ Interruptions

D.5 What to record

- Counts, durations, queue lengths, classification codes (U/Q/N/V/R).
- Environmental conditions and physical constraints.
- Descriptions of process and behaviour in general terms ("guard asked holder a question", not what was asked if it contained personal detail).
- Time-stamped exceptions and their outcomes.

D.6 What NOT to record

- Names, roll numbers, faces, photographs, video, audio.
- Card contents, vehicle registration numbers, visitor register contents.
- Identifying details of individual guards (name, badge, appearance).
- Conversations between guard and student beyond their general nature.
- Anything a guard or student asks you not to record.

D.7 Ethical considerations

- The gate is a public space and the observed behaviour is public; however, individuals have not consented, so the protocol collects only non-identifying aggregate data.
- Guards are informed in advance and may decline; observation of a guard who declines does not proceed.
- Observers must not create the impression of auditing or reporting on individual guards. The consent statement (Appendix C) is available to any guard who asks.
- If observers witness a security incident, they do not intervene and do not record identifying details; they note only "incident occurred, observation paused".

D.8 Field note sheet (per session)

Session: S_ Date: ____ Window: ____-____ Gate: ____ Observers: ____ / ____
 Guards present: ____ Weather/light: ____ Concurrent guard duties: ____
 Queue length at 2-min intervals: _____
 Timed persons (code, seconds): U 4, U 3, Q 22, N 15, ...
 Exceptions log:
 hh:mm type duration outcome note
 Process notes:
 Environment notes:
 Workarounds observed:
 Questions for interviews:

Timings are entered into `Data_Collection_Template.xlsx` (Appendix E), sheet "Observation".